VISVESVARAYA TECHNOLOGICAL UNIVERSITY
JNANA SANGAMA, BELAGAVI – 590 018, KARNATAKA, INDIA

INTERNSHIP REPORT

13/12/2024 – 02/04/2025

Research Internship/ Industry Internship

Submitted in partial fulfillment of the requirements for the award of
BACHELOR OF ENGINEERING
in

COMPUTER SCIENCE & ENGINEERING

Submitted By

Name	USN
SAMEEKSHA	4VP21CS078

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
VIVEKANANDA COLLEGE OF ENGINEERING & TECHNOLOGY
[A Unit of Vivekananda Vidyavardhaka Sangha, Puttur (R)]
Affiliated to Visvesvaraya Technological University and Approved by AICTE New Delhi & Govt. of Karnataka
Nehru Nagara, Puttur – 574 203, DK, Karnataka, India

May 2025

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CERTIFICATE

Certified that the Research Internship/ Industry Internship, carried out by **Ms. SAMEEKSHA,** bearing USN **4VP21CS078**, a bonafide student of **Vivekananda College of Engineering & Technology, Puttur**, in partial fulfillment for the award of **Bachelor of Engineering** in **Computer Science & Engineering** of the **Visvesvaraya Technological University, Belagavi** during the year 2024 – 25. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library.

The Internship report has been approved as it satisfies the academic requirements in respect of Research Internship/ Industry Internship (21INT82) prescribed for the said Degree.

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Signature of the Internship Guide
Prof. Thapaswini P S

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Signature of the Internship Coordinator
Prof. Deepthi M B

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Signature of the HOD
Prof. Pradeep Kumar K G

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Signature of the Principal
Dr. Mahesh Prasanna K

<u>EXTERNAL VIVA</u>

Name of the Examiners	Signature with date
1……………………………..............	....
2……………………………..............	....

EXECUTIVE SUMMARY & ACKNOWLEDGEMENT

I undertook a 15-week internship at **Rooman Technologies, Bengaluru**, from **December 13, 2024 to April 04, 2025.** I sincerely express my gratitude to Rooman Technologies, Bengaluru for providing me with the opportunity to undertake an internship at their esteemed organization. The experience has been immensely valuable, allowing me to enhance my technical skills and gain practical industry exposure. I am especially thankful to the entire team for their continuous support, guidance, and willingness to share their expertise. Their mentorship has greatly contributed to my learning and professional growth.

I would like to express my sincere gratitude to my internship guide **Prof. Thapaswini P S**, Assistant Professor, Department of CSE, for providing excellent guidance, encouragement and inspiration to complete the Internship.

I would like to extend my heartfelt gratitude to our internship coordinator, **Prof. Deepthi M B**, Assistant Professor, Department of CSE, for their exceptional guidance, unwavering support, and motivation throughout the Internship.

I would like to convey my heartfelt gratitude to **Prof. Pradeep Kumar K G**, Head of the Department of CSE, for his invaluable guidance and inspiration.

I sincerely thank our Principal, **Dr. Mahesh Prasanna K**, for offering all the required facilities and fostering a supportive environment to complete the Internship.

I extend my heartfelt thanks to the management, teaching, and non-teaching staff of the Computer Science & Engineering department for their invaluable support and assistance throughout the Internship.

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INTRODUCTION

1.1 About the Company

Rooman Technologies is a leading training provider dedicated to enhance employability, instilling confidence, and offering valuable certifications. From the beginning, the company focused on hiring the best trainers and developing innovative training programs aligned with global and national technological advancements. It has actively collaborated with the government to provide up skilling and reskilling opportunities, offering a wide range of government-supported vocational courses for graduates and students.

Wadhwani Foundation was established in 2000 by Dr. Romesh Wadhwani, the Wadhwani Foundation is a non-profit organization dedicated to fostering entrepreneurship, driving job creation, and enhancing skill development across emerging economies. The foundation empowers startups through mentorship and funding, supports small and medium enterprises (SMEs) in scaling operations, and delivers job-oriented training programs. Additionally, it champions AI-driven social impact initiatives aimed at addressing real-world challenges.

IBM (International Business Machines Corporation) was founded in 1911 as the Computing-Tabulating-Recording Company and rebranded as IBM in 1924, IBM is a globally recognized leader in technology and innovation. Headquartered in Armonk, New York, the company offers a wide range of products and services, including hardware, software, cloud computing, artificial intelligence, and IT consulting. IBM is renowned for ground breaking advancements such as Watson AI, quantum computing, IBM Cloud, enterprise mainframes, and cybersecurity solutions.

1.2 Company Profile

Company Name: Rooman Technologies

Main Office: 30, 12th Main Rd, 1st Stage, Rajajinagar, Bengaluru, Karnataka 560010

Email: online@rooman.net

Year Stand Up: 1999

Company Category: IT training, certification, and services

INTERNSHIP PROJECT OVERVIEW

CHAPTER 2

INTERNSHIP PROJECT OVERVIEW

2.1 Introduction to the Project Undertaken During the Internship

During my internship, I worked on a project titled “Automating Data Cleansing for Healthcare Records with NLP” which focused on improving the quality and reliability of unstructured textual data commonly found in the healthcare industry. This type of data, often collected from sources like clinical notes, electronic health records (EHRs), discharge summaries, medical prescriptions, and patient histories, is typically messy, incomplete, and inconsistent. It’s also highly prone to human errors, making it difficult to use directly in any meaningful analysis. Since manually cleaning such data is not only time-consuming but also prone to mistakes, the project aimed to create an automated solution using Natural Language Processing (NLP) techniques. The system was designed to detect and correct errors, fill in missing values, standardize medical terms using relevant ontologies, and remove duplicate or redundant information. By doing this, the data became cleaner, more structured, and ready for use in analytical tasks or machine learning models. This kind of preprocessing is especially important in sensitive areas like disease prediction, risk modeling, and personalized treatment planning. The overall goal was to make healthcare data more useful and actionable, allowing doctors and medical professionals to make better-informed decisions based on accurate, trustworthy information. This project not only made the data processing pipeline more efficient but also contributed to building a more data-driven and intelligent healthcare system.

2.2 System Requirements

System requirements outline the essential requirements for the successful development and deployment of the NLP-based automated healthcare data cleansing system. These are categorized into functional and non-functional requirements.

2.2.1 Functional Requirements

The Functional requirements define the specific behavior or functions of the system. For this project, the key functional requirements include:

- **Data Upload Module:** Allows users to upload raw healthcare datasets in CSV or similar formats.
- **Data Preprocessing Engine:** Automatic detects and handles missing, inconsistent data.

- **NLP-Based Standardization:** Applies NLP techniques to standardize medical terms and resolve textual ambiguities in clinical data.
- **Data Encoding:** Converts categorical variables into numerical format for machine learning and analysis.
- **Cleaned Data Output:** Generates and exports a cleaned, structured dataset ready for further processing.
- **User Interface:** Provides a simple interface for users to view data statistics.

2.2.2 Non-Functional Requirements

In contrast to functional requirements, which define what a system should do, non-functional requirements describe how the system should perform under various conditions. These may include aspects such as performance, scalability, reliability, usability, maintainability, and security. Non-functional requirements are crucial for ensuring the quality and user satisfaction of the system, and they often influence architectural decisions.

- **Reliability**
  The system should perform consistently under defined conditions without failure, ensuring dependable cleansing and processing results over time.

- **Performance**
  The Data preprocessing tasks such as missing value handling, encoding, and standardization should be completed within a reasonable time.

- **Usability**
  The system should be user-friendly for both technical users (data scientists) and non-technical users (healthcare staff), especially if deployed with a web interface.

- **Maintainability**
  The codebase should be modular and well-documented to allow for future enhancements or bug fixes.

- **Availability**
  The system should be accessible and operational whenever needed, with minimal downtime, especially when deployed in a real-time or healthcare-support environment.

2.3 Software Requirements

The project requires following software to run.

- Operating System: 64-bit Microsoft® Windows® 8/10 or MacOS 10
- Languages: Python, JavaScript, Node.js
- Software: Visual Studio Code

2.4 System Design

The system design outlines how different components of the automated data cleansing pipeline interact with each other. The architecture is modular, consisting of a data input module, a cleansing engine powered by NLP, and an output interface for exporting structured data. The key components communicate through well-defined workflows and APIs.

2.4.1 Flowchart

A flowchart, or process map, is one of the major tools of quality in project management, illustrating the step-by-step process of reaching given objectives in the best sequence of steps. By illustrating a sequence of actions, decisions, parallel processes and branches of loop, a flowchart provides a complete picture of the workflow of a task.

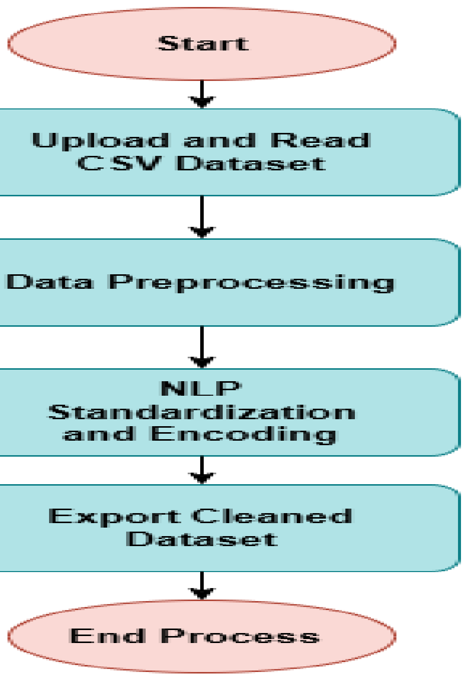

Figure 2.1: Flowchart of the System

The Figure 2.1 illustrates the complete workflow of the NLP-based healthcare data cleansing process. It begins with the initiation of the application, where the user uploads a raw healthcare dataset in CSV format. The system first performs preprocessing to handle missing values, null entries, and duplicate records. After this, it applies Natural Language Processing (NLP) techniques to standardize medical terminology, correct spelling errors, and resolve textual ambiguities. The next step involves encoding categorical variables into numerical format to prepare the data for machine learning or analytical use. Finally, the system exports the cleaned and structured dataset, marking the end of the process. This flow ensures that the data is consistent, accurate, and ready for advanced healthcare applications.

2.4.2 Use Case

A use case represents a collection of scenarios that illustrate the interactions between a source and a destination. The primary elements of a use case diagram are the use cases themselves and the actors involved. This diagram depicts the relationships between these elements. By utilizing a use case diagram, one can identify the various user types of a system and their corresponding use cases. It helps in understanding the functional requirements of the system from an end-user perspective.

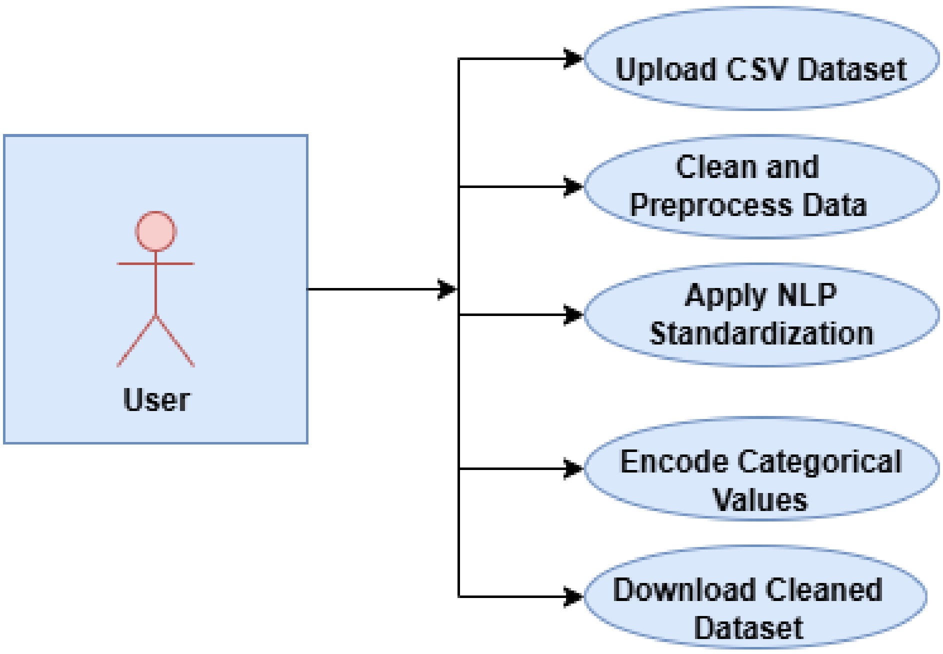

Figure 2.2: Use Case Diagram for the System

The Figure 2.2 illustrates the use case diagram for the NLP-based healthcare data cleansing system, highlighting the interaction between the user and the system. The user, who can be a data scientist or healthcare professional, performs several key actions through the interface. These include uploading a CSV dataset containing raw healthcare data, initiating the data cleaning and preprocessing phase where missing values, null entries, and duplicates are automatically handled, and applying NLP-based standardization to normalize medical terminology and correct textual inconsistencies. Additionally, the user can encode categorical variables into numerical formats suitable for machine learning applications, view a detailed cleansing report to compare the data before and after processing, and finally download the cleaned and structured dataset. This diagram provides a clear understanding of the functional requirements of the system by outlining the key operations a user can perform.

2.4.3 Sequence Diagram

A Sequence diagrams in Unified Modeling Language (UML) are a type of interaction diagram that illustrates how processes operate with one another and in what order. They are derived from Message Sequence Charts and are sometimes referred to as event diagrams, event scenarios, or timing diagrams. Sequence diagrams model the dynamic behavior of a system by showing how objects communicate through method calls and messages over time. They provide a clear visualization of the flow of control between components, helping to enhance understanding of system behavior and interactions.

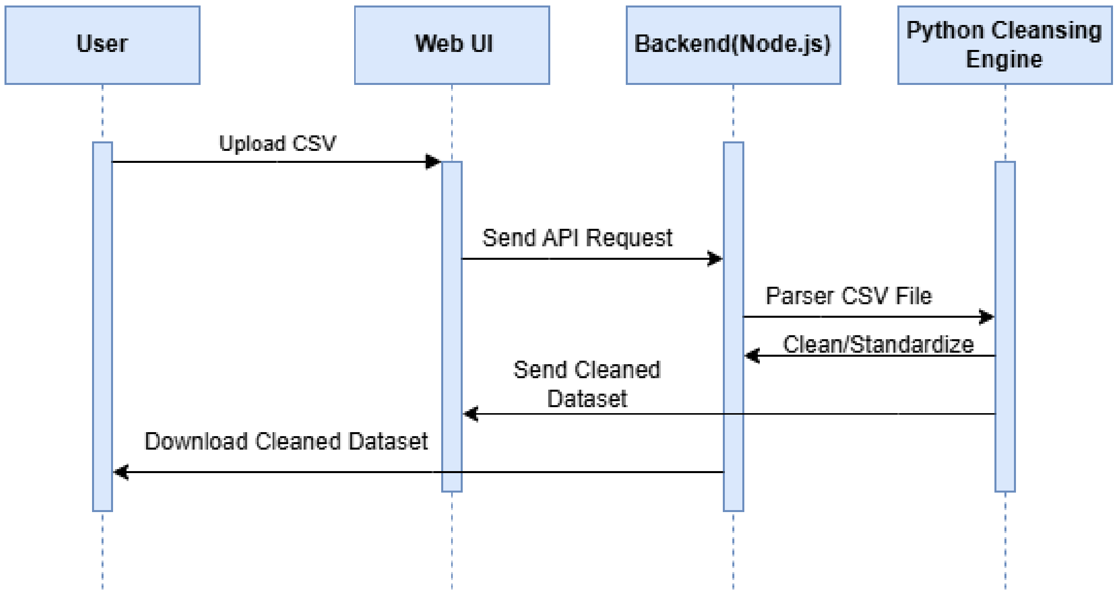

Figure 2.3: Sequence Diagram of the System

The Figure 2.3 illustrates the sequence of interactions in a data cleaning system involving four components: the user, web UI, backend (Node.js), and the Python cleansing engine. The process begins when the user uploads a CSV file through the Web UI, which then triggers an API request to the backend (Node.js). The backend receives the file and sends it to the Python Cleansing Engine for processing. The engine starts by parsing the CSV file to understand its structure and identify any issues. Once the data is parsed, the engine performs cleaning and standardization tasks, such as removing duplicates, correcting errors, or transforming the data into a consistent format. After the cleaning process is complete, the Python Cleansing Engine sends the cleaned dataset back to the backend, which in turn forwards it to the Web UI. Finally, the user can download the cleaned dataset from the Web UI, completing the process.

INTERNSHIP PROJECT IMPLEMENTATION

CHAPTER 3

INTERNSHIP PROJECT IMPLEMENTATION

3.1 Introduction

In the field of healthcare, data quality is essential for the success of predictive models and decision-making processes. However, healthcare datasets often contain inconsistencies, missing values, and noisy data. This project, "Automating Data Cleansing for Healthcare Records with NLP" aims to address these issues by automating the data preprocessing step using Natural Language Processing (NLP) and machine learning techniques. The primary objective is to develop a system that can cleanse raw healthcare data and prepare it for machine learning models or further analytics. This system ensures that the dataset is consistent, accurate, and usable for tasks like disease and symptom prediction, ultimately aiding in the development of reliable AI-powered diagnostics.

3.2 Implementation

The project implements a systematic data cleaning process that combines traditional preprocessing techniques with machine learning utilities to prepare healthcare datasets for further analysis or predictive modeling. The steps executed in the clean.py script are outlined below. This automated pipeline ensures consistency, reduces human error, and speeds up the data preparation phase. It lays a strong foundation for building accurate and reliable AI-driven healthcare applications.

3.2.1 Load Dataset

The data cleaning process begins with loading a raw healthcare dataset, typically stored in CSV format (e.g., Disease_symptom_and_patient_profile_dataset.csv). This dataset includes vital information such as patient symptoms, demographic profiles, and disease records. Accurate loading is essential to preserve the structure and integrity of the data. By initiating the process with this step, the system ensures a reliable input for all subsequent cleaning tasks. And tools are Python's pandas' library is used to load the dataset efficiently.

3.2.2 Detect Missing Values

After loading the data, the next critical step is to identify and report missing values column wise. This helps in determining which fields require cleaning or imputation. Detecting missing data at an early stage prevents issues in model training and ensures the completeness of the dataset. It also provides insights into data quality and highlights areas that may need special handling. The tool is used pandas.isnull() is used to detect missing values, and pandas.sum() aggregates the count of missing entries in each column.

3.2.3 Handle Missing Data

Handling missing data is essential to improve the overall quality and reliability of the dataset. Based on the type of data categorical or numerical the script applies suitable strategies to fill in the gaps. For categorical columns, missing values are filled with the mode, which helps maintain consistency and avoids introducing new, potentially invalid categories. For numerical columns, missing values are replaced with the mean, preserving the statistical distribution of the data and minimizing bias. This step ensures that the dataset remains complete and suitable for downstream analysis. The tools used are pandas.fillna() is utilized to impute missing values using mode and mean functions.

3.2.4 Encode Categorical Columns

Since machine learning models require numeric input, categorical data such as symptoms or disease names must be transformed into a numerical format. This transformation is accomplished through Label Encoding, a technique that assigns a unique integer to each distinct text value within a column. Converting textual information into numeric codes ensures compatibility with machine learning algorithms while preserving the categorical meaning of the data. This step is crucial for enabling predictive models to process and learn from the dataset effectively. The tools used are sklearn preprocessing. Label Encoder is used for transforming categorical text data into numeric codes.

3.2.5 Output Cleaned Dataset

Once the dataset has been cleansed and transformed, the final step involves exporting the cleaned data into a new CSV file named cleaned_dataset.csv. This refined dataset, now free from missing values and properly formatted, is ready for use in machine learning models or further analysis. To enhance user accessibility, the project incorporates a Flask-based web interface that allows users to download the cleaned file and view categorized patient records labeled as "Healthy" or "At Risk," offering meaningful insights based on the processed data. Tools Used: pandas.to_csv() is used to export the cleaned data, and Flask is utilized to display and visualize the output in a user-friendly format.

3.3 Snapshots

The Figure 3.1 displays a snapshot of the user interface developed for the project titled “Automating Data Cleansing for Healthcare Records with NLP.” This dashboard serves as a visual representation of the data cleansing and analysis process. At the top, the Quality Metrics section highlights key performance indicators such as Accuracy, Precision, Recall, and F1 Score, all showing values of 100%, indicating highly effective data preprocessing and model performance on the test data. Below this, the Outcome Variable Analysis section uses a pie chart to present the classification of patient records into "Healthy" (green) and "At Risk" (red) categories, allowing users to quickly assess the risk distribution. On the right side, the Live Healthcare Records table provides real-time visualization of patient data including attributes like Age, Gender, Fever, Cough, Fatigue, and Difficulty Breathing, along with the system-generated health Outcome and an Action button to flag any entry for further review. Overall, the dashboard reflects the project's objective to automate, standardize, and streamline healthcare data cleansing using NLP techniques, ensuring the processed data is both clean and analytically useful.

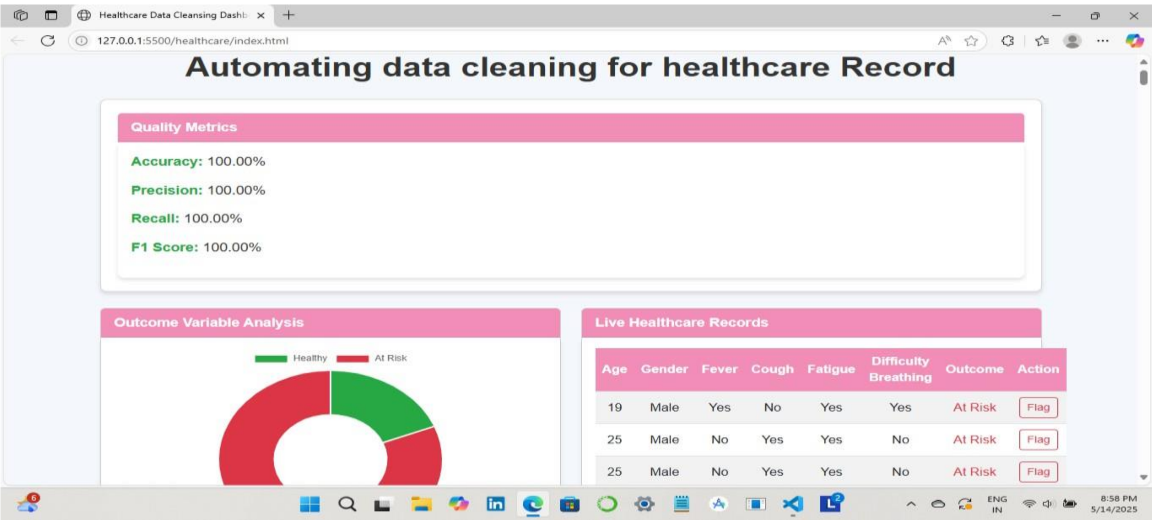

Figure 3.1: Dashboard Interface for Automated Healthcare Data Cleansing

ASSESSMENT & CONCLUSION

CHAPTER 4

ASSESSMENT & CONCLUSION

4.1 Assessment of Internship

The internship aimed to provide hands-on experience AI Data Quality Analyst internship, regular assessments were conducted to evaluate participants' understanding and progress in Python-based data analysis. These assessments were aligned with the weekly modules and practical labs, focusing on key topics such as string and numerical operations, array manipulation using NumPy, data handling with Pandas, and data visualization techniques. Both interim and final evaluations were designed to test theoretical knowledge, practical application, and problem-solving skills related to data cleaning, transformation, and quality analysis. These assessments helped reinforce the skills required to work with real-world datasets and ensured active engagement throughout the internship, preparing participants to contribute effectively to data-driven projects in various domains.

Week 1- Project Orientation: The first week covered foundational tools and concepts for modern development, including Agile and Scrum for project management, GitHub for version control and collaboration, and Design Thinking for user-focused innovation.

Week 2- Programming with Strings, NumPy, and Pandas: This week focused on core programming skills using Python. Learners explored string manipulation, efficient data handling with NumPy arrays, and data analysis with Pandas. Key topics included indexing, slicing, filtering, and reading and writing files laying the groundwork for effective data processing.

Week 3- Data Visualization and Outlier Detection: I explored data visualization using Matplotlib, Seaborn, and Plotly, creating various plots and dashboards. For outlier detection, I applied statistical methods (Z-score, IQR) and machine learning techniques (Isolation Forests, One-Class SVM, Autoencoders) across different domains.

Week 4- Foundational Data Processing: Covered NumPy operations for array math/CSV processing, data quality principles (accuracy, completeness), and practical Panda’s cleaning and analysis of housing data. Combined theory with Python implementation for real-world applications.

Week 5- Data Governance & Preprocessing: Covered data preprocessing (normalization, handling missing values), governance (GDPR/CCPA), and quality assessment using Pandas. Included practical labs on data cleaning and transformation for AI-ready datasets.

Week 6- Data Management: Covered data cleaning, integration (ETL/ELT), and quality control for big data. Explored specialized techniques for different data types and aggregation methods, balancing automation with human oversight for effective data preparation.

Week 7- Advanced Data Analysis & Modeling: Advanced from data cleaning to machine learning with the Titanic dataset. Covered feature engineering, classification models (Logistic Regression, Random Forest), and model evaluation. Focused on handling class imbalance and interpreting results for survival prediction.

Week 8- Data Quality & Feature Engineering: Covered advanced data cleaning (anomaly detection, consistency checks) and feature engineering (encoding, transformations). Used tools like Open Refine and Great Expectations for quality control, with PDCA for continuous improvement. Applied Python for practical data optimization.

Week 9- Data Pipelines & Talend for Data Quality: Covered the concept of data pipelines for quality assurance, explaining each stage from ingestion to monitoring and highlighting how quality checks are embedded throughout and Talend cloud is used for data quality management, highlighting its features like data profiling, cleansing, validation, and governance. It also discusses real-world applications and tools used in data quality management.

Week 10- Data Quality in Machine Learning & Monitoring it in AI Systems: Covered role of data quality in machine learning and responsibilities of data analyst, detailing how factors like accuracy, completeness, relevance affects model performance and techniques for ensuring data quality during training and strategies for monitoring it in deployment. It also presents real-world case studies from healthcare, e-commerce to illustrate the impact of poor versus high-quality data.

Week 11- Impact of Data Cleaning on Model Performance: Demonstrates the impact of data cleaning on model performance by comparing two scenarios: minimal data cleaning and proper data cleaning. It emphasizes that proper data cleaning, such as removing missing values and outliers, can significantly enhance model accuracy and reliability.

Week 12-13- Metrics in Machine Learning & Scikit-learn: Covered essential machine learning metrics for evaluating model performance across classification, regression, and clustering tasks, and introduces the Scikit-learn library and also understanding common machine learning models. It also includes a code implementation example using the Titanic dataset to demonstrate these data preparation steps.

Week 14-15- Final Project: Inclusive and Sustainable Workplace Practices: During the final phase, I applied all acquired knowledge to build a NLP-based automated cleansing for healthcare data from initial data quality assessment to model evaluation. I applied techniques for missing data handling, outlier detection and standardization, drawing from the data cleaning strategies covered in the course.

4.2 Conclusion

The internship provided a valuable and hands-on learning experience by integrating theoretical concepts with practical applications. The project, "Automating data Cleansing for Healthcare Record with NLP," effectively tackled a real-world problem by focusing on improving the consistency and reliability of healthcare datasets. By applying a structured data preprocessing pipeline spanning from missing value treatment to categorical encoding and data export the project demonstrated the crucial role of clean data in enabling accurate machine learning predictions. The use of Python, Flask, and machine learning libraries like sklearn facilitated robust backend development and visualization, while the web-based interface ensured user accessibility. This experience not only deepened technical proficiency in Natural Language Processing and data handling but also enhanced skills in teamwork, critical thinking, and project execution. Overall, the internship laid a solid foundation for pursuing future opportunities in data science, AI-driven healthcare solutions, and full-stack development.

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